Mathematical and Computational Methods

Exercise Sheet 6: Eigensystems

This sheet accompanies *Lecture 6 (Eigensystems)*. It exercises eigenvalues, eigenvectors, diagonalisation and their applications; nothing beyond Units 1–6 is needed.

How to use this sheet. Problems are graded by difficulty—$\star$ drill (routine fluency), $\star\star$ core (standard, tutorial level) and $\star\star\star$ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit6}"`) and recompile.

Conventions. All 2×2 work is meant to be done by hand; for larger matrices the eigenvalues or eigenvectors are given so the arithmetic stays clean. Use the trace/determinant identities $\sum_i \lambda_i = \operatorname{tr} \mathbf{A}$ and $\prod_i \lambda_i = \det \mathbf{A}$ to check your answers.

1 Warm-up drills

Exercise 1.1 *($\star$ drill – the eigen-relation)*. Let $\mathbf{A} = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$. Verify that $\mathbf{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ are eigenvectors, and state the eigenvalue of each. Check your two eigenvalues against $\operatorname{tr} \mathbf{A}$ and $\det \mathbf{A}$.

Exercise 1.2 *($\star$ drill – characteristic equation)*. Write down the characteristic equation of $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and solve it for the eigenvalues.

Exercise 1.3 *($\star$ drill – triangular matrices)*. Find, by inspection, the eigenvalues of $\mathbf{T} = \begin{pmatrix} 5 & 7 \\ 0 & -2 \end{pmatrix}$. Why can you read them straight off?

Exercise 1.4 *($\star$ drill – eigenvectors)*. For the matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ of Exercise 1.2, find an eigenvector for each eigenvalue.

Exercise 1.5 *($\star$ drill – trace/determinant check)*. The symmetric matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ has eigenvalues $1, 2, 4$. Compute $\operatorname{tr} \mathbf{A}$ and $\det \mathbf{A}$ and confirm they agree with the sum and product of the eigenvalues.

Exercise 1.6 *($\star$ drill – complex eigenvalues)*. (a) Find the eigenvalues of the quarter-turn $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. (b) Using $\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, write down the eigenvalues of the 60° rotation.

Exercise 1.7 *($\star$ drill – spot the structure)*. For each matrix say whether it is *symmetric*, *Hermitian*, both, or neither:

$$\text{(i)}\ \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \quad \text{(ii)}\ \begin{pmatrix} 2 & \mathrm{i} \\ -\mathrm{i} & 3 \end{pmatrix}, \quad \text{(iii)}\ \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}, \quad \text{(iv)}\ \begin{pmatrix} 1 & 2+\mathrm{i} \\ 2+\mathrm{i} & 4 \end{pmatrix}.$$

Exercise 1.8 *($\star$ drill – diagonalisable?)*. Decide, with a one-line reason, whether each matrix is diagonalisable:

$$\text{(a)}\ \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}, \qquad \text{(b)}\ \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}, \qquad \text{(c)}\ \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Exercise 1.9 *($\star$ drill – powers of a diagonal matrix)*. Let $\mathbf{D} = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$. Write down $\mathbf{D}^5$ and $\mathbf{D}^k$ for a general positive integer k.

Exercise 1.10 *($\star$ drill – assembling* **P** *and* **D***)*. A 2×2 matrix $\mathbf{A}$ has eigenpairs $\lambda_1 = 3$, $\mathbf{v}_1 = (1,1)^\mathsf{T}$ and $\lambda_2 = -2$, $\mathbf{v}_2 = (1,-1)^\mathsf{T}$. Write down matrices $\mathbf{P}$ and $\mathbf{D}$ with $\mathbf{A} = \mathbf{PDP}^{-1}$, and compute $\mathbf{P}^{-1}$.

Exercise 1.11 *($\star$ drill – valid transition matrices)*. A transition matrix has non-negative entries and *columns that sum to one*. Which of these qualify?

$$\text{(i)}\ \begin{pmatrix} 0.6 & 0.1 \\ 0.4 & 0.9 \end{pmatrix}, \quad \text{(ii)}\ \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.6 \end{pmatrix}, \quad \text{(iii)}\ \begin{pmatrix} 0.3 & 0.7 \\ 0.7 & 0.3 \end{pmatrix}.$$

Exercise 1.12 *($\star\star$ core – steady state)*. Find the steady-state distribution of the transition matrix $\mathbf{P} = \begin{pmatrix} 0.6 & 0.2 \\ 0.4 & 0.8 \end{pmatrix}$ (the eigenvector for $\lambda = 1$, scaled so its entries sum to one).

2 Tutorial problems

Problems marked ▶ form the *live tutorial core*: the set we work through together in the 2-hour class. The remaining problems in this section round out the same ideas and are best completed at home straight afterwards.

Exercise 2.1 *(▶ $\star\star$ core – a full 2×2 eigensystem)*. For $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ find the eigenvalues and an eigenvector for each, and verify the trace/determinant identities.

Exercise 2.2 *(▶ $\star\star$ core – diagonalise and find $\mathbf{A}^k$)*. Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. (a) Diagonalise $\mathbf{A} = \mathbf{PDP}^{-1}$. (b) Hence derive a closed form for $\mathbf{A}^k$, valid for every positive integer k. (c) Check your formula at $k = 1$.

Exercise 2.3 *($\star\star$ core – a Binet-type formula)*. A sequence satisfies $x_{n+1} = x_n + 2x_{n-1}$ with $x_0 = 0$, $x_1 = 1$ (giving $0, 1, 1, 3, 5, 11, \ldots$). (a) Write the step as $\binom{x_{n+1}}{x_n} = \mathbf{A}\binom{x_n}{x_{n-1}}$ and find $\mathbf{A}$ and its eigenvalues. (b) Decompose the start vector in the eigenbasis and obtain a closed form for x_n.

Exercise 2.4 *(▶ $\star\star$ core – a Hermitian 2×2 (physics))*. Let $\mathbf{H} = \begin{pmatrix} 3 & 2\mathrm{i} \\ -2\mathrm{i} & 3 \end{pmatrix}$. (a) Confirm $\mathbf{H}$ is Hermitian. (b) Find its eigenvalues (they should be real) and an eigenvector for each. (c) Verify the two eigenvectors are orthogonal under the conjugate inner product $\mathbf{u}^\dagger\mathbf{v} = \overline{u_1}v_1 + \overline{u_2}v_2$.

Exercise 2.5 *(▶ ⋆⋆ core – orthogonal diagonalisation of a symmetric 3×3)*. The symmetric matrix $\mathbf{A} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ -1 & -1 & 5 \end{pmatrix}$ has eigenvalues $\lambda = 2, 3, 6$, with eigenvectors $(1, -1, 0)^\mathsf{T}$, $(1, 1, 1)^\mathsf{T}$, $(1, 1, -2)^\mathsf{T}$ respectively. (a) Verify one eigenpair directly, say $\lambda = 6$. (b) Confirm the three eigenvectors are mutually orthogonal. (c) Normalise them to build an orthogonal matrix $\mathbf{Q}$ and write the spectral decomposition $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T}$.

Exercise 2.6 *(▶ ⋆⋆ core – a defective matrix)*. Show that $\mathbf{A} = \begin{pmatrix} 5 & -1 \\ 1 & 3 \end{pmatrix}$ is *not* diagonalisable.

Exercise 2.7 *(▶ ⋆⋆ core – a two-state Markov chain (data science))*. A streaming service tracks subscribers. Each month a *subscribed* user stays subscribed with probability 0.8 and lapses with 0.2; a *lapsed* user resubscribes with 0.4 and stays lapsed with 0.6. Let $\mathbf{x}_n = (\text{subscribed, lapsed})$ fractions. (a) Write the transition matrix $\mathbf{P}$ and find its eigenvalues; (b) find the steady-state distribution; (c) starting from everyone subscribed, $\mathbf{x}_0 = (1, 0)$, find a closed form for $\mathbf{x}_n$ and state the long-run subscribed fraction.

Exercise 2.8 *(▶ ⋆⋆ core – a spiral (physics))*. Find the eigenvalues and an eigenvector of $\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$. What does the modulus of the eigenvalues say about the orbit of $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$?

Exercise 2.9 *(⋆⋆ core – a covariance matrix (data science))*. A data set has (scaled) covariance matrix $\mathbf{A} = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}$. (a) Find its eigenvalues and eigenvectors. (b) Confirm the eigenvectors are orthogonal. (c) In PCA terms, along which direction does the data vary, and how much variance lies in the perpendicular direction?

Exercise 2.10 *(⋆⋆ core – true or false)*. State whether each is true or false, with a one-line justification.

(a) Every real 2×2 matrix has at least one real eigenvalue.

(b) If $\mathbf{A}$ is invertible, 0 is not an eigenvalue of $\mathbf{A}$.

(c) A 3×3 matrix with eigenvalues $2, 2, 5$ can never be diagonalised.

(d) Eigenvectors belonging to distinct eigenvalues are linearly independent.

(e) A real symmetric matrix can have a non-real eigenvalue.

(f) If λ is an eigenvalue of $\mathbf{A}$, then λ^2 is an eigenvalue of $\mathbf{A}^2$.

Exercise 2.11 *(⋆⋆ core – a short proof)*. Suppose $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Show that (a) $\mathbf{A}^2\mathbf{v} = \lambda^2\mathbf{v}$; (b) $(\mathbf{A} + c\mathbf{I})\mathbf{v} = (\lambda + c)\mathbf{v}$ for any scalar c; (c) if $\mathbf{A}$ is invertible then $\lambda \neq 0$ and $\mathbf{A}^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$.

Exercise 2.12 *(⋆⋆ core – spot the error)*. A student finds the eigenvalues of $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ like this: "$(2 - \lambda)^2 - 1 = 0$, so $(2 - \lambda)^2 = 1$, hence $2 - \lambda = 1$, giving $\lambda = 1$." Identify the mistake and give the correct eigenvalues.

Exercise 2.13 *(▶ ⋆⋆⋆ challenge – capstone proof: symmetric ⇒ real)*. Show that a real symmetric matrix $\mathbf{A} = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$ always has real eigenvalues. (This is the 2×2 case of the spectral theorem that underpins the orthogonal diagonalisation in Exercise 2.5.)

3 Further practice (home)

Exercise 3.1 *(⋆⋆ core – a battery of 2×2 eigensystems)***.** For each matrix find the eigenvalues and an eigenvector for each (allow complex values, and note any defective case):

$$\text{(a)} \begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}, \quad \text{(b)} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \quad \text{(c)} \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}, \quad \text{(d)} \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

Exercise 3.2 *(⋆⋆ core – a 3×3 with a convenient zero)***.** Find all eigenvalues and an eigenvector for each of $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$.

Exercise 3.3 *(⋆⋆ core – another matrix power)***.** Let $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$. Diagonalise it and find a closed form for $\mathbf{A}^n$.

Exercise 3.4 *(⋆⋆ core – another Hermitian matrix)***.** Let $\mathbf{H} = \begin{pmatrix} 1 & \mathrm{i} \\ -\mathrm{i} & 1 \end{pmatrix}$. Check it is Hermitian, find its eigenvalues and eigenvectors, and verify the eigenvectors are orthogonal under the conjugate inner product.

Exercise 3.5 *(⋆⋆ core – spot the error: a missing conjugate)***.** Continuing Exercise 2.4, a student checks whether the eigenvectors $\mathbf{v}_5 = (\mathrm{i}, 1)^\mathsf{T}$ and $\mathbf{v}_1 = (-\mathrm{i}, 1)^\mathsf{T}$ of the Hermitian matrix are orthogonal: "$\mathbf{v}_5^\mathsf{T}\mathbf{v}_1 = (\mathrm{i})(-\mathrm{i}) + (1)(1) = 1 + 1 = 2 \neq 0$, so they are *not* orthogonal." Identify the mistake and give the correct verdict.

Exercise 3.6 *(⋆⋆ core – rotation eigenvalues)***.** Write down the matrix of the rotation through $\theta = 120^\circ$ and state its eigenvalues in the form $\mathrm{e}^{\pm \mathrm{i}\theta}$ (and as $a \pm b\mathrm{i}$).

Exercise 3.7 *(⋆⋆ core – algebraic versus geometric multiplicity)***.** For each matrix, give the algebraic and geometric multiplicity of every eigenvalue, and say whether it is diagonalisable:

$$\text{(a)} \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad \text{(b)} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad \text{(c)} \begin{pmatrix} 4 & 2 \\ 0 & 1 \end{pmatrix}.$$

Exercise 3.8 *(⋆⋆ core – spectral decomposition as PCA)***.** The symmetric matrix $\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ (think of it as a covariance matrix) has eigenvalues $\lambda = 1, 2, 4$, with eigenvectors $(1, -1, 1)^\mathsf{T}$, $(1, 0, -1)^\mathsf{T}$, $(1, 2, 1)^\mathsf{T}$ respectively. (Mirroring Exercise 2.5, the eigenvectors are supplied so the work stays clean.) (a) Verify one eigenpair directly, say $\lambda = 4$. (b) Confirm the three eigenvectors are mutually orthogonal. (c) Normalise them to build an orthogonal matrix $\mathbf{Q}$ and write the spectral decomposition $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\mathsf{T}$. (d) In PCA language, which direction is the first principal axis and what fraction of the total variance does it carry?

Exercise 3.9 *(⋆⋆ core – principal component analysis (data science))***.** A two-feature data set has covariance matrix $\mathbf{C} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$ (each feature has variance 3; their covariance is 2). (a) Find the eigenvalues and eigenvectors of $\mathbf{C}$. (b) Which eigenvector is the first principal component? (c) What percentage of the total variance does it capture?

Exercise 3.10 *(★★ core – sketch: a spiral orbit)*. Revisit the complex-eigenvalue system of Exercise 2.8, $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$ with $\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$ and $\mathbf{x}_0 = (1, 0)^\top$. Compute $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, plot the four points, and sketch the orbit through them. Relate what you draw to the eigenvalues $\lambda = 1 \pm 2\mathrm{i}$.

Exercise 3.11 *(★★★ challenge – Lucas numbers and the golden ratio)*. The Lucas numbers obey $L_{n+1} = L_n + L_{n-1}$ with $L_0 = 2,\ L_1 = 1$ $(2, 1, 3, 4, 7, 11, \ldots)$. Using the matrix $\mathbf{A} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with eigenvalues the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$ (eigenvectors $(\varphi, 1)^\top, (\psi, 1)^\top$), show that $L_n = \varphi^n + \psi^n$.

Exercise 3.12 *(★★★ challenge – a population model (physics/biology))*. A two-stage population (juveniles j, adults a) updates by $\left(\begin{smallmatrix} j \\ a \end{smallmatrix}\right)_{n+1} = \mathbf{A}\left(\begin{smallmatrix} j \\ a \end{smallmatrix}\right)_n$ with $\mathbf{A} = \begin{pmatrix} 0 & 4 \\ 0.5 & 0 \end{pmatrix}$ (each adult yields 4 juveniles; half the juveniles survive to adulthood). Find the eigenvalues and hence the long-run growth factor per step and the stable juvenile-to-adult ratio.

Exercise 3.13 *(★★★ challenge – when does a chain settle?)*. A regular Markov chain settles to a unique steady state when every eigenvalue other than $\lambda = 1$ has modulus less than 1. For each transition matrix, find the eigenvalues and say whether the chain converges or oscillates:

$$\text{(a) } \mathbf{P} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}, \qquad \text{(b) } \mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Exercise 3.14 *(★★★ challenge – $\mathbf{A}$ and $\mathbf{A}^\top$)*. Show that a square matrix $\mathbf{A}$ and its transpose $\mathbf{A}^\top$ have the same eigenvalues.